



Red Hat



GuardDog AI Retail Use Case

Overview

In retail, lost business is often the largest portion of breach costs. It accounts for an average of \$1.59 million per incident, covering customer turnover and revenue loss from operational disruption.

The retail industry is estimated to account for about 6% of all global data breaches in 2025.

Third-party vendor compromises: 2025 has seen an increase in attacks targeting retailers' supply chains. In one report, 97% of top U.S. retailers experienced a third-party data breach within the last year, with some ransomware attacks originating from third-party systems.

Point-of-Sale (POS) systems: Approximately 50% of cyberattacks on retailers targeted POS systems.

IoT device exploitation: With more internet-connected devices in stores, attackers can exploit insecure sensors, smart shelves, and other devices to gain network access.

The Problem:

A large retail group with 17 locations is experiencing critical security challenges due to limited visibility and outdated security tools. Their current system only logs events after incidents occur, leading to a false sense of security and delayed threat response. They lack comprehensive awareness of their environment, underestimated the number of connected devices by over 700, and have gaps in their security infrastructure. These issues make them vulnerable, especially at remote sites, hybrid infrastructure, and un-agentable IoT devices.

The Solution:

GuardDog AI was deployed without re-architecture, delivering intelligent, proactive protection and frictionless adoption across the client's environment, easing the burden on the small IT and security team. The platform addressed existing tool gaps by providing real-time network monitoring and automatic threat detection and containment across all locations — including remote sites, data centers, and cloud assets. Deployment uncovered the full asset inventory, identifying devices previously unknown, and enabled rapid, sub-second responses without increasing management complexity.

The Outcome

The deployment of GuardDog AI revolutionized cybersecurity for the retail client by providing comprehensive attack surface management across all store locations, distribution centers, cloud environments, and legacy systems. Its universal compatibility across retail infrastructure—without endpoint agents—enabled seamless security coverage across dispersed environments. The platform's AI-powered sensors rapidly contained threats in less than a second and generated zero false positives, transforming security operations from reactive to proactive.

During the initial rollout, GuardDog AI identified nearly 9,000 vulnerabilities and discovered 2,500 active devices—over 700 more than the client previously believed existed. This full visibility provided the retail chain with real-time automated threat containment, eliminating the need for manual intervention and significantly speeding up response times. Its scalable, multi-tenant architecture supported the client's multiple store locations and distribution centers, while also providing opportunities for future expansion.

By enhancing existing security investments and delivering autonomous, real-time AI detection and containment, GuardDog AI enabled the retail client's security team to focus on strategic remediation efforts, boosting their cybersecurity resilience. These improvements contributed to downstream benefits such as reduced cyber insurance premiums, increased operational efficiency, and stronger customer trust through improved data security and protection across all retail environments.

Retail Challenges – The Problem in Detail

Retail organizations typically operate multiple geographically dispersed locations, wholesale centers, and administrative offices – each with its own network segments, IoT devices, and customer-facing systems.

This distributed footprint creates consistent challenges:

Limited visibility across locations. IT teams often lack a unified view of their infrastructure, making it hard to know exactly what assets are connected.

Small security teams and toil. Many run with only one or two dedicated IT/cybersecurity staff, making proactive threat hunting and management difficult.

Complex, hybrid environments. Mixes of on-premises servers, cloud services, and remote access points increase the attack surface. Customer-facing network segments, Wi-Fi and retail systems, created additional ingress points for attackers.

High dependence on IoT and un-agentable devices. Point-of-sale terminals, phones, customer Wi-Fi networks, service bay systems, printers, and vehicle diagnostic tools cannot run traditional endpoint agents, leaving them exposed.

Gaps in real-time detection. Threats could move laterally within the network before being noticed. Complex segmentation with trusted and untrusted traffic made it easier for misconfigurations to introduce vulnerabilities.

Why traditional tools were insufficient.

The client had tools in place but relied on them primarily for log parsing and after-the-fact analysis, not real-time containment, and thus a false sense of security. Although the security tools had relevant capabilities, teams avoided using them due to configuration complexity and high management overhead. Additionally, tools were siloed, leaving gaps between detection, investigation, and containment – with no unified view. Further, existing tools were unable to protect un-agentable devices, where endpoint-centric solutions did not cover IoT, PoE, POS or transient mobile devices moving between networks.

The result was an environment where vulnerabilities persisted unnoticed, attackers could exploit internal blind spots and already stretched IT teams lacked the means to respond instantly – all conditions that GuardDog AI was purpose-built to address.

The Solution (i.e. How GuardDog AI was deployed)

Architecture

GuardDog Complete was deployed as an agentless, container-based solution running on Red Hat Enterprise Linux 9.5 servers. This sidecar-style architecture required no endpoint installations, enabling protection for both agentable and un-agentable devices. By operating at the network layer, GuardDog AI monitored and contained threats in real-time without impacting existing systems or workflows.

The deployment targeted the client's full hybrid environment, including:

- On-premises infrastructure at retail, wholesale and service center locations
- Cloud workloads (unspecified public cloud usage by the client)
- IoT and PoE devices such as POS systems, diagnostic tools, and wholesale/customer Wi-Fi endpoints
- Data center assets and inter-site connections

Although initial rollout covered only a subset of the ~17 locations, the design supported expansion to all sites, the central data center, and cloud environments.

Deployment Method

For this engagement, GuardDog AI worked directly with the client's internal IT team (two dedicated staff: one networking lead and one cybersecurity lead) rather than through an MSSP or OEM partner. The deployment process included:

- Installing and configuring the operating system on the designated hardware
- Pre-deploying GuardDog Complete on each server
- Delivering a plug-and-play experience so the client could bring systems online with minimal intervention
- Providing visibility and automated threat containment from day one

Typical deployments of this architecture can be completed within a single working day per site, with faster scaling possible using imaging tools for multiple servers in parallel.

Key Features

GuardDog is an ideal security fabric for automotive retail: automated, agentless, scalable, and designed to strengthen visibility and real-time protection across complex, distributed environments.

Coverage for un-agentable devices (PoE, IoT, POS, and diagnostic tools). GuardDog operates agentlessly, it protects devices that can't install traditional endpoint software i.e. retail displays, diagnostic systems, and POS terminals.

Hybrid and distributed environment visibility. With seamless deployment across on-prem, cloud, and edge, GuardDog gives a unified security view across dealership branches and central infrastructure.

Containerized Protection for all Devices. Designed to require minimal setup and management, perfect for businesses with limited IT and cybersecurity staff.

Real-time detection and isolation on customer-facing networks. Automatically isolates threats from customer Wi-Fi or guest networks without customer-facing disruption.

The Results

Vulnerability Discovery: Identified nearly 9,000 vulnerabilities during initial deployment.

Asset Visibility: Discovered 2,500 active devices, 700+ more than the client believed were in use.

Team Efficiency: Enabled a 2-person IT/security team to monitor and contain threats across 17 locations without adding staff.

Reduced Incident Response Time and Toil. From hours or days (log parsing/manual triage) to milliseconds for detection and sub-second containment. Cut dwell time dramatically, limiting breach scope and potential data loss.

Compliance and Insurance Benefits. Supports PCI DSS (POS device security) by providing real-time monitoring and isolation. Enables audit-ready reporting for SOC 2, ISO 27001, and data protection laws (e.g., CCPA), and strengthens security posture for cyber insurance applications, potentially lowering premiums.

Total Toil and Cost Savings *

- Avoided breach remediation costs – which in the automotive retail sector, breaches average \$3.88-4.88Million
- Reduced overtime and contractor spend by removing the need for constant manual investigation – an estimated 20 – 30% OpEx savings on security operations.
- Possible 5 – 15% reduction in annual cyber insurance premiums with documented controls and real-time containment.
- IT/Engineering productivity gains by eliminating the need for daily manual log review and device inventory updates – a savings of 10–15 hours per week for small teams. This freed security staff to focus on strategic initiatives rather than firefighting false positives, and improved coordination between the network and security roles by consolidating visibility into a single interface.

* Estimates based on industry averages.

Testimonial

"GuardDog AI gave us visibility we didn't know we were missing – finding over 700 devices we didn't even know existed and nearly 9,000 vulnerabilities in the process. What used to take hours or days to investigate is now detected and contained in milliseconds, without adding to our IT toil and workload. It's like having a 24/7 security team that never sleeps, protecting every location, every device, in real-time."